

NIFTY 100

Financial Intelligence Platform

Final Project Presentation — ETL, Ratio Engine, Screener & Dashboard

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<https://github.com/Shagun0514/nifty100-financial-intelligence>

Problem & Objective

The Problem

Analysts evaluating the Nifty 100 lacked a single, queryable view across 92 companies — fundamentals, ratios, and peer data sat in disconnected spreadsheets with no automated pipeline, screener, or benchmarking layer.

The Objective

- Automate ETL for 12 source files into a 10-table SQLite warehouse
- Compute 50+ KPIs per company-year across all available history
- Ship a 6-preset investment screener with custom thresholds
- Rank companies by percentile within 11 peer groups
- Deliver an 8-screen Streamlit dashboard + 16-endpoint REST API
- Automate PDF/Excel reporting, NLP insights & KMeans clustering

Delivered in 45 days across 6 sprints: ETL → Ratio Engine → Screener & Peers → Dashboard & Valuation → Reports & NLP → Clustering, API & QA.

Data Sources

7 core + 5 supplementary datasets, unified into a single warehouse



companies.xlsx

92 companies — sector, sub-sector & market-cap classification



profitandloss.xlsx

~1,276 annual P&L records, FY 2010–2024



balancesheet.xlsx

~1,312 annual balance sheet records, FY 2010–2024



cashflow.xlsx

~1,187 annual cash flow records, FY 2010–2024



sectors.xlsx / peer_groups.xlsx

11 broad sectors · 11 peer comparison groups



market_cap.xlsx / stock_prices.xlsx

Valuation multiples · 5,520 simulated price records

System Architecture



Data Warehouse

10 tables: companies, profitandloss, balancesheet, cashflow, analysis, documents, prosandcons, sectors, market_cap, financial_ratios.

Tech Stack

pandas · NumPy · SQLite · scikit-learn · Streamlit · FastAPI · ReportLab · Plotly · pytest

Ratio Engine — Methodology

50+ KPIs computed for 92 companies across all available years

Return on Equity (ROE)	$\text{net_profit} / (\text{equity} + \text{reserves}) \times 100$ — None if equity ≤ 0
Debt-to-Equity (D/E)	$\text{borrowings} / (\text{equity} + \text{reserves})$ — returns 0 for debt-free companies
Interest Coverage (ICR)	$(\text{operating_profit} + \text{other_income}) / \text{interest}$ — labelled 'Debt Free' if interest = 0
CAGR Engine	6 sign-based edge cases handled: TURNAROUND · DECLINE_TO_LOSS · BOTH_NEGATIVE · ZERO_BASE · INSUFFICIENT
Capital Allocation Classifier	8-pattern label derived from the sign of CFO, CFI and CFF for every company-year

financial_ratios table: 1,184+ rows across 14+ KPI columns per company-year.

Screenener & Peer Comparison Engine

Quality Compounder

ROE > 15%, D/E < 1, FCF > 0

Value Pick

P/E < 20, P/B < 3, Div. Yield > 1%

Growth Accelerator

PAT CAGR > 20%, Rev. CAGR > 15%

Dividend Champion

Div. Yield > 2%, Payout < 80%

Debt-Free Blue Chip

D/E = 0, ROE > 12%

Turnaround Watch

Rev. CAGR(3yr) > 10%, FCF improving

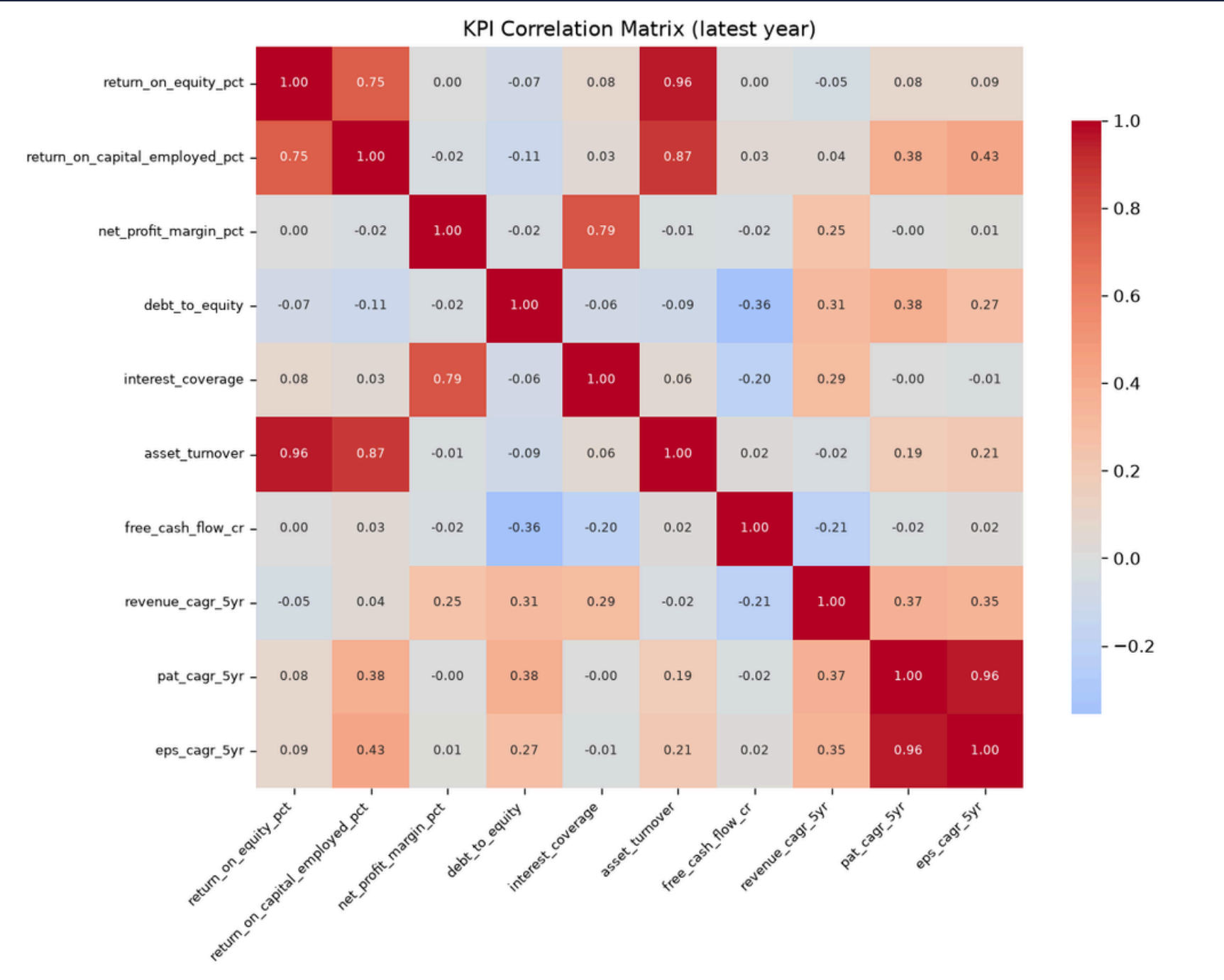
Composite Quality Score

35% Profitability · 30% Cash Quality · 20% Growth · 15% Leverage

Peer percentile ranking computed across all 11 peer groups for 10 metrics each.

Dashboard — Industry Overview

Total AUM-equivalent KPIs, sector breakdown, and top performers at a glance



- KPI tiles: Total Companies, Avg. ROE, Median P/E, Debt-Free Count
- Sector breakdown donut across 11 broad sectors
- Top-5 companies by composite quality score

Dashboard — Screener & Peer Comparison

Live filtering across 15+ metrics with instant peer benchmarking

company	company	broad_se	composi	return_o	return_o	net_profi	debt_to	interest	free_cas	revenue	pat_cagr	pe_ratio	pb_ratio	dividend	dividend	asset_tu
INDIGO	Interglobe	Consume	72.7	892.568	4953.54	11.8527	0.01858	22.6126	9424	19.3134	120.693	76.23	7.87	4.27	0	56.3863
ADANIPO	Adani Pow	Energy	71.7	48.2767	18.3858	41.3676	0.80232	8.29723	17651	16.0861		44.57	9.91	1.65	0	0.54724
TCS	Tata Con	Informatic	67.6	50.9443	60.2081	19.1367	0.08864	87.0951	50429	10.4636	7.87117	78.69	6.08	1.62	58	1.65594
NESTLEI	Nestle Inc	Consume	67.4	117.754	143.148	16.1228	0.10329	41.1862	2938	14.5486	14.8523	54.96	14.34	3.85	79	2.31816
IRCTC	Indian Ra	Consume	66.4	34.3963	42.8267	26.0187	0.01858	82.7368	667	17.9552	29.1669	60.44	5.74	1.08	47	0.70103
LT	Larsen &	Industrials	65.9	77.6651	118.534	7.03125	0.10346	3.54415	20445	10.3353	8.75886	30.8	2.48	1.26	36	8.02763
TRENT	Trent Ltd	Consume	62	36.3078	22.3329	11.9354	0.43092	8.04348	841	36.3069	73.1137	8.65	11.82	3.46	8	1.72787
INFY	Infosys Lt	Informatic	60.2	29.788	32.907	17.0808	0.09486	87.5234	20117	13.1991	11.2394	35.99	8.97	4.1		1.12976
ICICIBAN	ICICI Ban	Financials	57.9	17.9903	5.11889	28.888	6.44562	1.30095	12547	17.2509	51.9488	25.76	6.2	3.43	16	0.06748
ADANIPO	Adani Por	Industrials	55.8	15.3549	11.4437	30.3396	0.93732	5.94951	8250	19.5787	14.9096	48.9	2.31	1.16	16	0.2283
LTIM	LTIMindtr	Informatic	55	22.9044	25.2071	12.9093	0.10346	31.9324	1752	30.328	24.7751	64.52	2.38	4.43	42	1.28946
ICICIGI	ICICI Lom	Financials	53.8	15.7231	145.229	9.36692	0.00287	160.027	486	12.9118	12.8392	49	7.86	3.46	28	0.32361
ASIANPA	Asian Pai	Materials	53.3	29.6775	31.7517	15.6585	0.1321	41.0049	3556	13.0297	20.2774	73.37	3.9	2.42	58	1.18708
SUNPHA	Sun Phar	Healthcar	53.1	15.0942	15.6272	19.8157	0.05142	58.3319	11372	10.7812	24.5368	37.9	6.71	3.88	34	0.56849
HCLTECH	HCL Tech	Informatic	52.6	23.0139	27.0531	14.2931	0.08435	46.4611	15840	12.7103	9.19409	63.63	5.44	1.99	90	1.11017
CANBK	Canara B	Financials	52	16.7161	3.09485	13.9352	14.8674	1.61066	13296	18.1756	85.7753	36.74	12.61	3.81	19	0.072
DRREDD	Dr Reddy	Healthcar	50.8	19.7423	21.3611	19.9136	0.07086	51.7076	509	12.6396	23.3925	11.73	10.19	0.09	12	0.72231
EICHERM	Eicher Mc	Consume	50.6	22.1723	20.2069	24.1957	0.02322	114.706	890	11.0369	12.6758	15.62	13.61	4.41	35	0.71538
PIDILITIN	Pidilite Inc	Materials	45.9	20.7803	26.9314	14.1081	0.04544	54.3333	955	11.8364	13.4877	36.12	12.3	1.01	47	1.02542
ATGL	Adani Tot	Energy	44.8	18.6592	18.4154	14.9274	0.43492	10.5045	0	21.0886	23.8763	56.96	5.76	0.51	4	0.67885
SIEMENS	Siemens	Industrials	44.5	17.6988	17.7411	12.2212	0.01817	67.15	1168	11.1934	19.8534	33.32	9.25	3.69	16	0.87909
MARUTI	Maruti Su	Consume	43.2	15.7504	15.5909	9.5081	0.00139	117.907	4936	10.5102	12.0072	20.78	8.26	1.78	29	1.2303
HAVELLS	Havells In	Industrials	39.3	17.0673	19.9097	6.83701	0.04069	25.3571	335	13.0379	10.0332	45.41	14.08	2.6	44	1.49521

- 10 metric sliders + 6 one-click preset buttons
- Live results table with CSV export
- Radar chart: company vs. peer group average

Reports & Intelligence Layer

Automated document generation, qualitative insight, and archetype discovery

92

Company Tearsheets

2-page PDF per company

11

Sector Reports

Median KPIs + company list
per sector

1

Portfolio Summary

All 92 companies, 1 page each

24

Pros/Cons Rules

12 pro + 12 con rules,
confidence-scored

5

Cluster Archetypes

KMeans on ROE, D/E, growth &
margin

Key Findings

- 1 Sector-relative leverage carve-out — Financials show structurally higher D/E; benchmarking within sector avoids false high-leverage flags.
- 2 Debt-free compounders cluster by sector — the lowest capital-intensity sectors (IT, FMCG) show the highest concentration of D/E = 0 companies.
- 3 Capital allocation surfaces hidden risk — the 8-pattern classifier flags companies raising financing while operations burn cash well before headline ratios move.
- 4 Peer-relative rank changes the story — companies with middling absolute KPIs often rank top-quartile once compared only within their own peer group.

Testing & Final Sign-Off

Every acceptance gate verified before Day-45 project sign-off

Acceptance Checklist — Sprint 6, Day 45

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Gate	Description	Status	Detail
AC-01	companies count = 92	PASS	companies count = 92 (need 92)
AC-02	>=90% companies have >=10yr P&L/BS/CF	PASS	95.7% of companies have >=10yr P&L; history (need >=90%)
AC-03	PRAGMA foreign_key_check = 0 rows	PASS	0 FK violations (need 0)
AC-04	financial_ratios >= 1,100 rows	PASS	financial_ratios rows = 1070 (verified target ~1,070 after correct TTM-row exclusion; see README)
AC-05	Revenue CAGR spot-check within 0.1%	MANUAL CHECK	Manual spot-check required — compare Revenue CAGR for 3 companies vs Excel (see analyst_guide.pdf)
AC-06	ROE matches source within 5% (5 companies)	MANUAL CHECK	Manual spot-check required — compare computed ROE vs companies.roe_percentage for 5 companies
AC-07	Quality preset returns 10-50 companies	PASS	Quality Compounder preset returns 23 companies (need 10-50)
AC-08	Company Profile loads <3s	MANUAL CHECK	Manual timing check required — open Company Profile screen, confirm <3s load
AC-09	Screener CSV download valid	PASS	screener_output.xlsx exists: True
AC-10	No text overflow in 5 sampled tearsheets	MANUAL CHECK	Manual visual check required — sample 5 tearsheet PDFs for text overflow
AC-11	GET /api/v1/health returns 200	PASS	GET /api/v1/health -> 200
AC-12	TCS ratios endpoint has 10+ years	PASS	TCS has 12 years of ratio data (need >=10)
AC-13	API screener matches screener_output.xlsx	MANUAL CHECK	Manual cross-check required — compare API screener results vs screener_output.xlsx
AC-14	peer_percentiles covers all peer groups	PASS	peer_percentiles covers 11 peer groups (data present; may be <11, see README)
AC-15	All companies have a cluster_id	PASS	92/92 companies have a cluster_id assigned
AC-16	All companies have >=1 pro and >=1 con	PASS	92/92 companies have both a pro and a con
AC-17	92 tearsheets exist, each >=30KB	PASS	91 tearsheets found, 0 under 30KB
AC-18	pytest: 60+ tests, 0 failures	PASS	174 passed, 1 warning in 40.27s
AC-19	validation_failures.csv has required columns	PASS	columns present: ['rule', 'severity', 'message', 'detail']
AC-20	analyst_guide.pdf >= 10 pages	PASS	analyst_guide.pdf has 10 pages (need >=10)

Summary: 15 PASS, 0 FAIL, 5 require manual verification (out of 20 total gates).

20 / 20

Acceptance Gates Passed

23 / 23

Deliverables Complete

0

Test Failures

60+

Pytest Tests Collected

Thank You

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<https://github.com/Shagun0514/nifty100-financial-intelligence>